How-To Library

Informatica

Tuning Enterprise Data Catalog Performance in 10.4.0

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Abstract

This article provides information about tuning Enterprise Data Catalog performance. Tuning Enterprise Data Catalog performance involves tuning parameters for metadata ingestion, ingestion database, search, and tuning data profiling. The performance of Enterprise Data Catalog depends on the size of data that needs to be processed.

Supported Versions

- Enterprise Data Catalog 10.4.0

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Overview

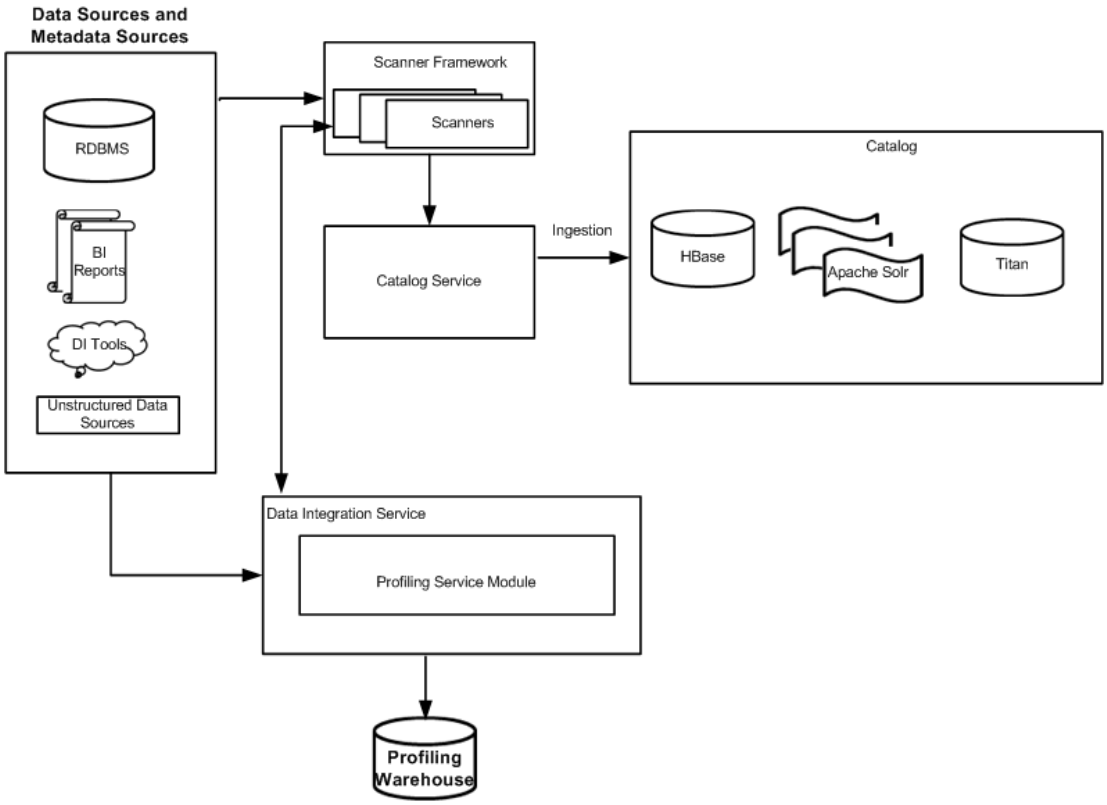
Tuning Enterprise Data Catalog Performance involves tuning the performance parameters at different stages of extracting metadata, profiling, identifying similar columns, tuning the catalog, and searching for assets in the catalog.

You can extract metadata from external sources, such as databases, data warehouses, business glossaries, data integration resources, or business intelligence reports. The performance of Enterprise Data Catalog depends on the size of data being processed. Enterprise Data Catalog classifies data sizes into three categories based on the size of data. Based on the data size, you can configure custom properties in Informatica Administrator to assign predefined parameter values for metadata ingestion, Apache HBase database tuning, and search. Alternatively, you can also individually configure the values for the performance tuning parameters based on your requirement. The profiling tuning involves tuning parameters for the Data Integration Service and the profiling warehouse database properties.

The metadata extraction, storage, search and related operations in an organization include the following phases:

1. Storing the extracted metadata into a catalog for ease of search and retrieval. A catalog represents a centralized repository for storing metadata extracted from different sources. This phase is referred to as the metadata ingestion phase. Enterprise Data Catalog uses Apache HBase as the database for ingesting data.
2. Validate the quality of data with data profiling.
3. Search for the related data assets in the catalog. Enterprise Data Catalog uses Apache Solr to search for data assets.

The following image shows the various stages at which you can tune the performance of Enterprise Data Catalog:



The following table lists the stages and the corresponding parameters that you can tune to optimise Enterprise Data Catalog performance:

Stage	Description	Parameter List
Data Sources and Metadata Sources	Extraction of metadata from different sources.	Maximum # of Connections that denotes the maximum number of connections or resources.
Scanner Framework	A framework that runs scanners and manages a registry of available scanners in Enterprise Data Catalog. A scanner is a plug-in component of Enterprise Data Catalog that extracts specific metadata from external data sources.	- Maximum # of Connections - Metadata extraction scanner memory parameter
Enterprise Data Catalog Performance Tuning: Parameters for Tuning Metadata Ingestion	The process of ingesting the extracted metadata to the catalog.	- Apache Spark parameters - Enterprise Data Catalog custom ingestion options
Enterprise Data Catalog Performance Tuning: Parameters for Tuning Apache HBase, Apache Titan, and Solr	The indexed archive used by Enterprise Data Catalog to store all the extracted metadata for search and retrieval.	- Apache HBase parameters - Apache Titan parameters - Apache Solr parameters
Data Integration Service for Profiling	An application service that performs data integration tasks for Enterprise Data Catalog and external applications.	- Profiling Warehouse Database - Maximum Profile Execution Pool Size - Maximum Execution Pool Size - Maximum Concurrent Columns - Maximum Column Heap Size - Maximum # of Connections
Profile Configuration in Data Integration Service: Profiling Warehouse Database Properties	The datastore where the profiling information is stored before the profiling information is moved to the catalog.	- Maximum Patterns - Maximum Profile Execution Pool Size
Similarity	Similarity discovers similar columns in the source data within an enterprise.	- sparkJobCount - sparkExecutorCount - sparkExecutorCoreCount - sparkExecutorMemory

Minimum System Requirements for a Hadoop Node in the Cluster

The system requirements listed in the table are for a Hadoop node in the cluster. Make sure that you allocate the required additional system requirements for the operating system, processes, and other applications running on the node. The values listed are recommendations by Informatica for improving performance. You can increase the values based on your requirements.

System Requirements

The following table lists the system recommendations for deploying Enterprise Data Catalog in the embedded Hadoop cluster:

System Requirement	Minimum Value
CPU	Eight physical cores with medium clock speed. **Note:** For optimum performance, you must use cores with a minimum clock speed of 2.6 GHz.
Memory	24 GB of unused memory available for use.
Hard disk drive	4 to 6 or more high-capacity SATA hard disk drives with 200 GB to 1 TB capacity and 7200 RPM for each disk. The hard disks must be configured as a Just a Bunch Of Disks (JBOD). Do not use any logical volume manager tools on the hard disk drives.
File system	ext4 file system.
Network	Bonded Gigabit Ethernet or 10 Gigabit Ethernet between the nodes in the cluster.

Note: If you upgraded from Enterprise Data Catalog version 10.2, follow the system requirements listed in the 10.2 version of this article available in the Informatica Knowledge Base.

Enterprise Data Catalog Sizing Recommendations

Based on the size of data, you must add additional memory and CPU cores to tune the performance of Enterprise Data Catalog. You must also note the minimum number of cores that are required to deploy supported data sizes.

The size of data depends on the number of assets (objects) or the number of datastores. Assets include the sum total of databases, schemas, columns, data domains, reports, mappings and so on in the deployment environment. A datastore represents a repository that stores and manages all the data generated in an enterprise. Datastores include data from applications, files, and databases. Determine if you need to change the default data size for the Enterprise Data Catalog installation. Enterprise Data Catalog has Low, Medium, and High data sizes or load type that you can configure in Informatica Administrator using custom properties. Data sizes are classified based on the amount of metadata to process and the number of nodes used to process metadata.

Note: The minimum recommended load type for production deployment is Medium. When the number of supported concurrent users for a load type increases, it is recommended that you move to the next higher load type.

After installation, you can switch the data size from a lower data size to a higher data size. For example, if you had selected a low data size during installation, you can change the data size to medium or high after installation. However, if you had selected a higher data size value during installation, for example, high, you cannot change the data size to a lower data size, such as medium or low after installation.

Note: Make sure that you restart and index the Catalog Service if you switch the data size after you install Enterprise Data Catalog.

Low

Low represents one million assets or 30-40 datastores. The maximum user concurrency to access Enterprise Data Catalog is 10 users and the recommended resource concurrency is 1 medium data size resource or 2 low data size resources.

System Requirements

The following table lists the recommended minimum system requirements for application services and infrastructure for low load type:

Component	Infrastructure*	Data Integration Service[+]	Profiling Warehouse	Single Node Hadoop Cluster
CPU Cores	8	8	4	8
Memory	16 GB	16 GB	16 GB	24 GB
Storage	200 GB	20 GB	50 GB	120 GB

*Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, and Informatica Cluster Service.

[+]The hardware sizing requirement listed is for a use case that includes profiling.
Note: You can run the Data Integration Service on the same node as the domain or on a separate node. If the Data Integration Service used for profiling co-exists on the Infrastructure node, the total hardware sizing is the sum of Data Integration Service and Infrastructure hardware components.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the hardware requirements listed in the table by 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 5 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

Note: The Catalog Service heap size is 2 GB. The Maximum Heap Size for Data Integration Service is 2 GB.

Service Level Agreement

The Service Level Agreement (SLA) to process metadata extraction and profiling for a low data size is approximately two business days. The SLA calculation is for a basic RDBMS resource such as Oracle.

Calculation

When you calculate the system requirements, make sure that you add the system requirements for infrastructure and the Hadoop cluster node, and as required for profiling and Catalog Service tuning.

Note: The complexity of transformations present in the extracted metadata affect the memory required for lineage. Ensure that you increase the memory for infrastructure as required.

Medium

Medium represents 20 million assets or 200-400 datastores. The maximum user concurrency to access Enterprise Data Catalog is 50 users and the recommended resource concurrency is 3 medium data size resources. A medium data size resource requires a minimum of three nodes to run the Hadoop cluster.

You can choose to use the minimum system requirements or system requirements for a high volume source as necessary. The system requirement tables list the components and values for infrastructure, Data Integration Service, and profiling warehouse.

Minimum System Requirements

The following table lists the recommended minimum system requirements for application services and infrastructure for medium load type:

Component	Infrastructure*	Data Integration Service+	Profiling Warehouse	Hadoop Cluster System Requirements per Node	Hadoop Cluster System Requirements (A minimum of three nodes required.)
CPU Cores	24	32	8	8	24
Memory	32 GB	64 GB	32 GB	24 GB	72 GB
Storage	200 GB	100 GB	200 GB	600 GB	1.8 TB

* Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, and Informatica Cluster Service.

+The hardware sizing requirement listed is for a use case that includes profiling.
Note: You can run the Data Integration Service on the same node as the domain or on a separate node. If the Data Integration Service used for profiling co-exists on the Infrastructure node, the total hardware sizing is the sum of Data Integration Service and Infrastructure hardware components.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the hardware requirements listed in the table by 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 20 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

Note: The Catalog Service heap size is 4 GB. The Maximum Heap Size for Data Integration Service is 10 GB.

Service Level Agreement

The following table describes the Service Level Agreement (SLA) to process metadata extraction and profiling for medium load type:

Process	SLA
Metadata extraction	Approximately one business week to accumulate 20 million assets from multiple resources. The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter.
Profiling	Approximately one business week in addition to metadata extraction SLA. The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling.

Calculation

When you calculate the system requirements, make sure that you add the system requirements for infrastructure and three Hadoop cluster nodes, and as required for profiling and Catalog Service tuning.

Recommended System Requirements for High Volume Sources

A high volume source might include a single large resource or large number of resources. As a general guideline, if the object count in a resource exceeds the recommended limit for medium scanner type, the resource is considered to be a high volume source. For more information about the object count, see the "Metadata Extraction Scanner Memory Parameters" on page 18 topic.

System Requirements

The following table lists the system requirements for application services and infrastructure for high volume sources for the medium load type:

Component	Infrastructure*	Data Integration Service[1]	Profiling Warehouse	Hadoop Cluster System Requirements per Node	Hadoop Cluster System Requirements (A minimum of three nodes required.)
CPU Cores	24	64	8	12	36
Memory	32 GB	128 GB	32 GB	48 GB	144 GB
Storage	200 GB	100 GB	200 GB	600 GB	1.8 TB

* Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, and Informatica Cluster Service.

[1] Indicates that you can use the specified memory for Data Integration Service or use two Data Integration Service nodes of 32 cores and 64 GB each. The hardware sizing requirement listed is for a use case that includes profiling.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the hardware requirements listed in the table by 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 20 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

For more information about high volume sources, see the "High Volume Sources" on page 11 topic.

Service Level Agreement

The following table describes the SLA to process metadata extraction and profiling for medium load type for high volume sources:

Process	SLA
Metadata extraction	Approximately three business days to accumulate 20 million assets from multiple resources. The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter.
Profiling	Approximately four business days in addition to metadata extraction SLA. The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling.

The throughput or performance depends on the number of columns in the table, file size, file type, complexity of mapping, size of the reports, file content, and number of data domains for profiling.

Recommendations

You can consider the following recommendations when you want to optimize performance for high volume sources in the medium load type object limit:

- To increase the ingestion throughput and utilize the recommended hardware, increase the Spark executor count **--num-executors** in the *custom.properties.medium* file located in the `<INFA_HOME>/services/CatalogService/Binaries` folder and restart the Catalog Service.
  The increase in each Spark executor count uses three cores and 5 GB of memory.
- To increase the profiling throughput and use the recommended hardware effectively, see the "setConcurrencyParameters Script" on page 29 topic to tune the parameters as necessary.

High

High represents 50 million assets or 500-1000 datastores. The maximum user concurrency to access Enterprise Data Catalog is 100 and the recommended resource concurrency is 5 medium data size resources. A high data size requires a minimum of six nodes to run the Hadoop cluster.

You can choose to use the minimum system requirements or system requirements for a high volume source as necessary. The system requirement tables list the components and values for infrastructure, Data Integration Service, and profiling warehouse.

Minimum System Requirements

The following table lists the recommended minimum system requirements for application services and infrastructure for high load type:

Component	Infrastructure*	Data Integration Service†	Profiling Warehouse	Hadoop Cluster System Requirements per Node	Hadoop Cluster System Requirements (A minimum of six nodes required.)
CPU Cores	48	32	16	8	48
Memory	64 GB	64 GB	64 GB	24 GB	144 GB
Storage	300 GB	500 GB	500 GB	2 TB	12 TB

* Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, and Informatica Cluster Service.

†The hardware sizing requirement listed is for a use case that includes profiling.
Note: You can run the Data Integration Service on the same node as the domain or on a separate node. If the Data Integration Service used for profiling co-exists on the Infrastructure node, the total hardware sizing is the sum of Data Integration Service and Infrastructure hardware components.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the hardware requirements listed in the table by 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 100 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

Note: The Catalog Service heap size is 9 GB. The Maximum Heap Size for Data Integration Service is 20 GB.

Service Level Agreement

The following table describes the SLA to process metadata extraction and profiling for high load type:

Process	SLA
Metadata extraction	Approximately two business weeks to accumulate 50 million assets from multiple resources. The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter.
Profiling	Approximately two business weeks in addition to metadata extraction SLA. The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling.

Calculation

When you calculate the system requirements, make sure that you add the system requirements for infrastructure and six Hadoop cluster nodes, and as required for profiling and Catalog Service tuning. For better throughput with additional hardware capacity, see the performance tuning section for each component in this article.

Recommended System Requirements for High Volume Sources

A high volume source might include a single large resource or large number of resources. For more information about the object count, see the "Metadata Extraction Scanner Memory Parameters" on page 18 topic.

System Requirements

The following table lists the recommended system requirements for application services and infrastructure for high volume sources for the high load type:

Component	Infrastructure*	Data Integration Service[1]	Profiling Warehouse	Hadoop Cluster System Requirements per Node	Hadoop Cluster System Requirements (A minimum of six nodes required.)
CPU Cores	48	64	32	12	72
Memory	64 GB	128 GB	64 GB	48 GB	288 GB
Storage	300 GB	500 GB	500 GB	2 TB	12 TB

* Includes Informatica domain services, Model Repository Service, Catalog Service, Content Management Service, and Informatica Cluster Service.

[1] Indicates that you can use the specified memory for Data Integration Service, or use two Data Integration Service nodes of 32 cores and 64 GB each. The hardware sizing requirement listed is for a use case that includes profiling.

For instances hosted on the cloud, the number of vCores/vCPUs greatly impacts the overall system performance. It is recommended that you use compute-optimized virtual machines and increase the hardware requirements listed in the table by 2x. Also, it is recommended that you have premium SSD storage for better throughput and higher IOPS.

The `/tmp` and `/var` directories must have a minimum of 100 GB of free disk space.

For high volume use cases, it is recommended that you increase the storage of the `/tmp` directory as the volume of source data grows. The storage can be further increased for very large volumes of source data and cloud based sources. Consider the size of the largest files and parallel profile jobs and accordingly increase the storage allocated for the `/tmp` directory.

For more information about high volume sources, see the "High Volume Sources" on page 11 topic.

Service Level Agreement

The following table describes the SLA to process metadata extraction and profiling for high volume sources for the high load type:

Process	SLA
Metadata extraction	Approximately one business week to accumulate 50 million assets from multiple resources. The SLA calculation is for a basic RDBMS resource such as Oracle or PowerCenter.
Profiling	Approximately two business weeks in addition to metadata extraction SLA. The SLA calculation is for a basic RDBMS resource, such as Oracle, with 10K sampling rows for profiling.

The throughput or performance depends on the number of columns in the table, file size, file type, complexity of mapping, size of the reports, file content, and number of data domains for profiling.

Recommendations

You can consider the following recommendations when you want to optimize performance for high volume sources in the high load type object limit:

- To increase the ingestion throughput and utilize the recommended hardware, increase the Spark executor count **--num-executors** in the *custom.properties.high* file located in the `<INFA_HOME>/services/CatalogService/Binaries` folder and restart the Catalog Service.
  The increase in each Spark executor count uses three cores and 5 GB of memory.
- To increase the profiling throughput and use the recommended hardware effectively, see the "setConcurrencyParameters Script" on page 29 topic to tune the parameters as necessary.

High Volume Sources

A high volume source might include a single large resource or large number of resources. As a general guideline, if the object count in a resource exceeds the recommended limit for medium scanner type, the resource is considered to be a high volume source. For more information about the object count, see the "Metadata Extraction Scanner Memory Parameters" on page 18 topic.

Recommendations to Optimize the Performance

You can tune the components to optimize the performance when you scan a high volume source, or when one or more of the following use cases is true:

- Re-indexing a large catalog.
- Re-assigning the connection assignment parameters.
- Deleting a large resource.
- Scanning a large PowerCenter repository.
- Scanning a large-volume business intelligence report.
- Scanning an Oracle source with large volume public synonym.
- Increasing upgrade throughput when you move to a higher version.
- Profiling millions of unstructured files.

To optimize the performance, you can consider the following recommendations:

- Split a large resource into multiple resources, if possible.
- Increase the disk space by 10 GB for every additional 1 million assets.

- When an ingestion job remains in the same stage and does not move to the next stage, cancel the job and increase the value of the **ingest.max.ingest.facts.int** and **--executor-memory 3G** properties in the *custom.properties.high* file located in the `<INFA_HOME>/services/CatalogService/Binaries` directory.

  It is recommended that you increase the property values to three times the current value. For example, if the configured values are *ingest.max.ingest.facts.int=600000* and *--executor-memory 3G*, then set the properties to *ingest.max.ingest.facts.int=1800000* and *--executor-memory 9G*.

- When a single large resource exceeds the **High** scanner memory option limit, increase the **LdmCustomOptions.scanner.memory.high** value as necessary in the **Informatica Administrator** > **Manage** > **Services and Nodes** > **Catalog Service** > **Custom Properties** section and restart the Catalog Service.

  For example, when you want to run complex mappings based on large resources, large reports, or large unstructured files, or if you want to include all the schemas in a resource scan, then increase the custom property value.

- When you want to run a profile on millions of unstructured files, increase the node process heap size to 2 GB.

  To increase the node process heap size, perform the following steps:

  1. Open the **infaservice.sh** file located in the `<INFA_HOME>/tomcat/bin/` directory.
  2. Increase the value of the *INFA_JAVA_OPTS* parameter from **INFA_JAVA_OPTS="-Xmx1024m"** to **INFA_JAVA_OPTS="-Xmx2048m"**.
  3. Save the file.
  4. Restart Informatica domain.

- When the number of files and tables in a resource exceed 100,000, increase the Model Repository Service heap size to 2GB.

  To increase the heap size, navigate to **Informatica Adminstrator** > **Services and Nodes** > **Model Repository Service** > **Properties** > **Advanced Properties** section and set **Maximum Heap Size** to **2048m**.

- Before you run high volume business intelligence resources where the report export file size exceeds 50MB, increase the **--executor-memory** parameter value to 10 GB.

- Before you purge or delete a high volume resource in Catalog Administrator, increase the **--executor-memory** parameter value to 6 GB.

  To configure the **--executor-memory** parameter to 6 GB, navigate to `<INFA_HOME>/services/CatalogService/Binaries` directory, edit the *custom.properties.<load_type>* file, enter **--executor-memory 6G**, and save the file.

- Before you re-run a high volume resource in an upgraded instance, you might need to increase the **--executor-memory** parameter based on the data size of the resource.

Note: Restart the Catalog Service after you add or change a custom property.

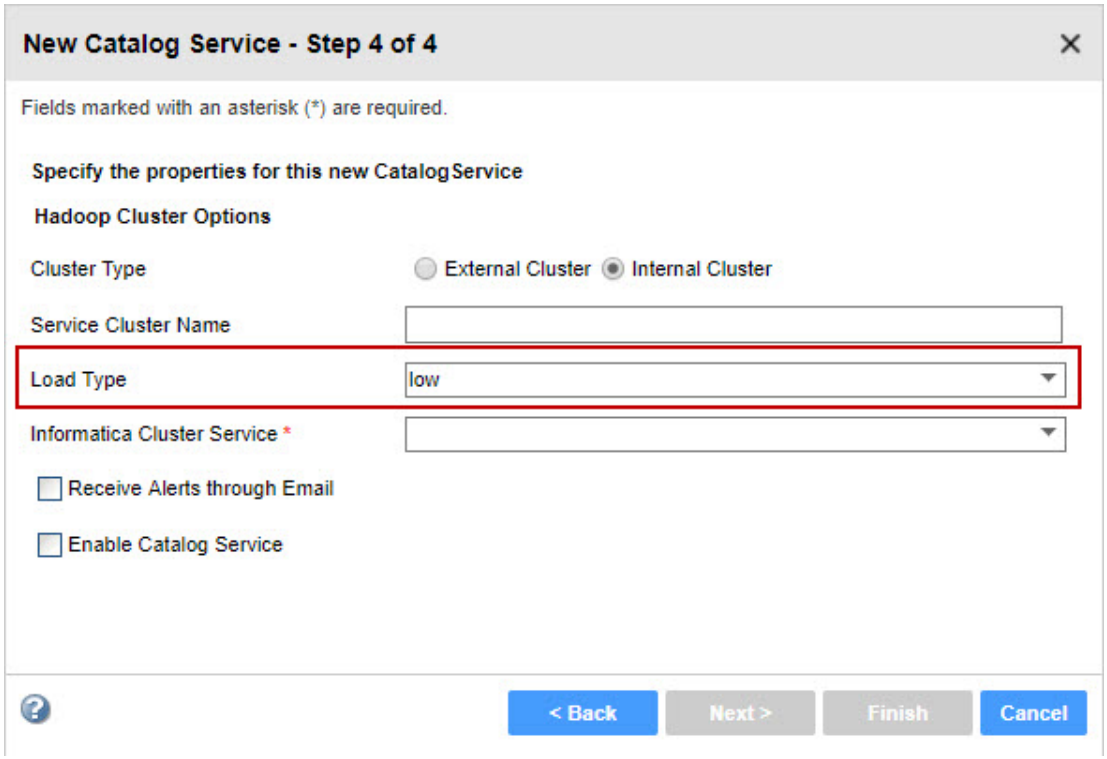
Tuning Performance Based on the Size of Data

Enterprise Data Catalog includes predefined values for the performance tuning parameters based on the volume of supported data sizes. You can specify the required data size when you create the Catalog Service. After you specify the data size, Enterprise Data Catalog uses the predefined values associated with the data size to configure the performance tuning parameters. You can also tune each parameter based on your requirements.

You can view and edit the predefined values for the performance tuning parameters in the `custom.properties.<loadtype>` files. The `custom.properties.low`, `custom.properties.medium`, and `custom.properties.high` files are located in the `<INFA_HOME>/services/CatalogService/Binaries` folder.

When you create a Catalog Service, you can choose low, medium, or high in the **New Catalog Service - Step 4 of 4** > **Load Type** option to specify the data size in the catalog.

The following sample image shows the **New Catalog Service - Step 4 of 4** dialog box:



See the *Informatica Enterprise Data Catalog Installation and Configuration Guide* for more information about creating the Catalog Service.

Predefined Parameter Values for Data Sizes

You can modify the metadata ingestion, Apache HBase database tuning, and Apache Solr parameters in the `custom.properties.<loadtype>` files to tune the performance of Enterprise Data Catalog. Tuning the parameters based on the size of data helps to improve the performance of Enterprise Data Catalog. Data sizes are classified based on the amount of metadata that Enterprise Data Catalog processes and the number of nodes in the Hadoop cluster. You can calculate the size of data based on the total number of objects, such as tables, views, columns, schemas, and business intelligence resources.

When you increase the memory configuration of any component, for example, ingestion, it is recommended that you maintain a buffer of 30% of the actual memory required for the component. For example, if a component requires 100 MB of memory, you must increase the memory configuration to 130 MB for that component.

Ingestion Parameters

Ingestion parameters includes Apache Spark and Enterprise Data Catalog custom options.

Apache Spark Parameters

The following table lists the predefined ingestion parameters for *LdmSparkProperties* property based on data sizes:

Apache Spark Parameters	Low	Medium	High
driver-memory*	1024m	1536m	2048m
executor-memory*	3G	3G	3G

Apache Spark Parameters	Low	Medium	High
num-executors	1	3	6
executor-cores[1]	2	2	2
conf spark.storage.memoryFraction	0.4	0.3	0.3
conf spark.yarn.executor.memoryOverhead	1280	1280	1280
conf spark.yarn.driver.memoryOverhead	384	384	512
conf spark.kryoserializer.buffer.max	128m	128m	128m
* When you increase the value for this parameter, it is recommended that you increase the maximum memory allocation in YARN. for a container. Failing to increase the memory allocation might result in YARN shutting down the applications. [1] For external clusters, when you increase the value for this parameter, it is recommended that you increase the maximum number of cores in YARN, for a container.			

Note: To linearly improve the ingestion performance, increase the num-executors parameter for LdmSparkProperties, and increase the CPU and memory resources.

Enterprise Data Catalog Ingestion Options Parameters

The following table lists the predefined ingestion parameters for *ExtraOptions* property based on data sizes:

Enterprise Data Catalog Ingestion Options Parameters	Low	Medium	High
ingest.batch.delay.ms	120000	120000	120000
ingest.xdoc.batch.size.int	1000	1000	1000
ingest.max.ingest.xdocs.int	4000	10000	20000
ingest.max.ingest.facts.int	100000	300000	600000
ingest.enable.propagation	true	true	true
ingest.index.partitions.int	96	96	96
ingest.partitions.int	20	60	120
janus.ids.num-partitions	1	3	6
janus.cluster.max-partitions	64	64	64
janus.storage.hbase.region-count	8	8	8
ingest.propagation.partitions.int	96	96	96
ingest.sleep.time.ms	45000	90000	120000
ingest.xdoc.memory.buffer.bytes	104857600	314572800	524288000

Enterprise Data Catalog Ingestion Options Parameters	Low	Medium	High
ingest.hbase.table.regions	8	24	48
hclient.phoenix.query.timeoutMs	600000	600000	1800000

Apache HBase Parameters

Apache HBase parameters includes HBaseSiteConfiguration parameters, HBaseRegionServerProperties parameters, HBaseMasterProperties parameters, and HBaseSliderAppMaster parameters.

HBase Site Configuration Parameters

The following table lists the predefined *HbaseSiteConfiguration* parameters based on data sizes:

HBase Site Configuration Parameters	Low	Medium	High
hbase_master_heapsize*	512m	768m	1024m
hbase_regionserver_heapsize*	3072m	3072m	3072m
hbase.master.handler.count	30	30	600
hbase.regionserver.handler.count	50	50	100
hbase.hstore.blockingStoreFiles	1000	1000	1000
hbase.hregion.majorcompaction	86400000	86400000	86400000
hbase.hregion.majorcompaction.jitter	1	1	1
* When you increase the value for this parameter, it is recommended that you increase the maximum memory allocation in YARN. for a container. Failing to increase the memory allocation might result in YARN shutting down the applications.			

HBase Region Server Properties

The following table lists the predefined *HbaseRegionServerProperties* parameters based on data sizes:

HBase Region Server Parameters	Low	Medium	High
yarn.component.instances[2]	1	3	6
yarn.memory	3992	3992	3992
yarn.vcores[1]	1	1	1
[1] For external clusters, when you increase the value for this parameter, it is recommended that you increase the maximum number of cores in YARN, for a container. [2] Before increasing this parameter, you must add the required number of nodes to the cluster.			

HBase Master Properties

The following table lists the predefined *HBaseMasterProperties* parameters based on data sizes:

HBase Master Properties Parameters	Low	Medium	High
yarn.component.instances[2]	1	1	1
yarn.memory	768	1024	1228
yarn.vcores[1]	1	1	1
[1] For external clusters, when you increase the value for this parameter, it is recommended that you increase the maximum number of cores in YARN, for a container. [2] Before increasing this parameter, you must add the required number of nodes to the cluster.			

HBase Slider App Master Properties

The following table lists the predefined *HbaseSliderAppMasterProperties* parameters based on data sizes:

HBase Slider App Master Properties Parameters	Low	Medium	High
jvm.heapsize	128M	128M	128M
yarn.component.instances[2]	1	1	1
yarn.memory	512	512	512
yarn.vcores[1]	1	1	1
[1] For external clusters, when you increase the value for this parameter, it is recommended that you increase the maximum number of cores in YARN, for a container. [2] Before increasing this parameter, you must add the required number of nodes to the cluster.			

Apache Solr Parameters

Apache Solr parameters includes parameters for Solr slider app master properties and Solr node properties to tune the performance of the Apache Solr for search operations.

Solr Slider App Master Properties

The following table lists the predefined *SolrSliderAppMasterProperties* parameters based on data sizes:

Solr Slider App Master Properties Parameters	Low	Medium	High
jvm.heapsize	128M	128M	128M
yarn.component.instances[2]	1	1	1
yarn.memory	512	512	512

Solr Slider App Master Properties Parameters	Low	Medium	High
yarn.vcores[1]	1	1	1
[1] For external clusters, when you increase the value for this parameter, it is recommended that you increase the maximum number of cores in YARN, for a container. [2] Before increasing this parameter, you must add the required number of nodes to the cluster.			

Solr Node Properties

The following table lists the predefined *SolrNodeProperties* parameters based on data sizes:

Custom Options for Solr Node Properties Parameters	Low	Medium	High
yarn.component.instances[2]	1	3	6
yarn.memory	2560	2560	2560
yarn.vcores[1]	1	1	1
xmx_val*	2048m	2048m	2048m
xms_val	512m	512m	512m
solr.replicationFactor	1	1	1
solr.numShards	1	3	6
* When you increase the value for this parameter, it is recommended that you increase the maximum memory allocation in YARN. for a container. Failing to increase the memory allocation might result in YARN shutting down the applications. [1] For external clusters, when you increase the value for this parameter, it is recommended that you increase the maximum number of cores in YARN, for a container. [2] Before increasing this parameter, you must add the required number of nodes to the cluster.			

Metadata Extraction Scanner Memory Parameters

Depending on the size of source data for a resource, you can use one of the following parameters to configure the memory requirements for the scanner to extract metadata:

Parameter	Memory Option*	Supported Object Count for Default Memory[1]	Default Memory[2]
LdmCustomOptions.scanner.memory.low	low	Up to 1000 tables Up to 2000 files Up to 100 mappings/reports Up to 100,000 objects Up to a maximum file size of 5 MB for unstructured files	1 GB
LdmCustomOptions.scanner.memory.medium	medium	Up to 10,000 tables Up to 20,000 files Up to 1000 mappings/reports Up to 1M objects Up to a maximum file size of 10 MB for unstructured files	4 GB
LdmCustomOptions.scanner.memory.high	high	Up to 100,000 tables Up to 1,000,000 files Up to 20,000 Mappings/Reports Up to 10M objects Up to a maximum file size of 100 MB for unstructured files	12 GB
* Indicates the memory allocated to scanner process heap size. [1] Depends on the number of columns in the table, file size, file type, complexity of the mapping, size of the report, and object content. [2] Indicates the default values configured for the scanner based on the data size.			

Increase the value of the **LdmCustomOptions.scanner.memory.high** custom property as necessary or configure the *-Dscanner.memory.high=24576* JVM option when one of the following conditions is true:

- You want to run complex mappings based on large resources, large reports, or large unstructured files.
- You want to include all the schemas in a resource scan.
- A single large resource exceeds the High scanner memory option limit.

To increase the value of the **LdmCustomOptions.scanner.memory.high** custom property, perform the following steps:

1. In Informatica Administrator, navigate to the **Manage** > **Services and Nodes** > **Catalog Service** > **Custom Properties** section.
2. Edit the property, enter the required value, and save the property.
3. Restart the Catalog Service after you change the custom property.

To configure the *-Dscanner.memory.high=24576* JVM option, perform the following steps:

1. In Catalog Administrator, edit the resource.

2. In the **Metadata Load Settings** > **Advanced Properties** section, configure the following options:
   - Memory. Choose **High**.
   - JVM Options. Enter **-Dscanner.memory.high=24576**.
3. Save and run the resource.

Note:

- You must select the number of concurrent scanners based on the memory type and the available resources in the cluster.
- You must increase the Informatica PowerCenter scanner memory based on the complexity of the mappings.

Tuning Configuration Parameters for Resources

You can configure parameters to optimize the performance of resources.

Cloud Resources

You can tune the performance parameters for cloud resources such as Azure Data Lake Storage and Google Cloud Storage resources.

To increase the source metadata extraction throughput, tune the **-DfileProcessorCount** JVM option. By default, this option is set to **2** for all file-based resources.

You can configure the **-DfileProcessorCount** parameter based on the number of cores required for a node in the cluster as shown in the following table:

Number Of Cores Required for a Node In The Cluster	JVM Option and Value
8	DfileProcessorCount=8
16	DfileProcessorCount=16

To add, edit, or delete a JVM option, navigate to the **Metadata Load Settings** tab of the resource in Catalog Administrator.

Tuning Performance for Similarity

The following table lists the default values for parameters associated with column similarity, based on the size of data:

Parameter	Low	Medium	High
sparkJobCount	1	1	2
sparkExecutorCount	1	1	1
sparkExecutorCoreCount	1	2	2
sparkExecutorMemory (in GB)	4	5	5

Note the following points before you tune the parameters based on your requirements:

- Similarity performance scales linearly when you increase the *sparkExecutorCount* and the *sparkExecutorCoreCount* parameters.

- Scale-up performance during similarity profile run depends on the values you configure for the total parallel mappings including native profiling and similarity profiling.
- To increase the similarity throughput and utilize the additional hardware, increase the *sparkJobCount* and *SparkExecutorCount* parameter as necessary. The increase in each Spark executor count uses three cores and 7 GB of memory.

Enabling Spark Dynamic Resource Allocation

In the Administrator tool, perform the following steps to enable dynamic resource allocation for Spark applications for an existing cluster:

1. Disable the Catalog Service.
2. In the **Catalog Service** > **Custom Properties** section, set the `LdmCustomOptions.ingest.spark.dynamic.allocation.enable` custom property to `true` to enable Spark dynamic allocation.
3. Add or modify the following parameters in the `yarn-site.xml` file present on every host:
   - Set the `yarn.nodemanager.aux-services` parameter to `spark_shuffle` value.
   - Set the `yarn.nodemanager.aux-services.spark_shuffle.class` parameter to `org.apache.spark.network.yarn.YarnShuffleService` value.

   In the master node, the `yarn-site.xml` file is located in the `/var/lib/ambari-server/resources/stacks/HDP/<hadoop_version_number>/services/YARN/configuration/` folder.

   In the other nodes, the `yarn-site.xml` file is located in the `/var/lib/ambari-agent/cache/stacks/HDP/<hadoop_version_number>/services/YARN/configuration/` folder.
4. Copy `spark-<version>-yarn-shuffle.jar` to the `/usr/hdp/current/hadoop-yarn-nodemanager/lib/` directory of every host where node managers are running.

   Note: You can get the `spark-<version>-yarn-shuffle.jar` file from the`/lib` directory located inside the Spark binary tar file or you can download the .jar file. Make sure that the version of the .jar file is the same as the Spark version.
5. Restart the node managers.
6. Enable the Catalog Service.

Note: These steps do not apply if you have deployed Enterprise Data Catalog on an embedded cluster.

Improving Search Performance Based on Data Size

To improve the search performance in Enterprise Data Catalog, you can configure the custom property for Solr deployment options and Solr node properties based on the data size. After you change the custom options, you need to restart the Catalog Service to implement the changes.

LdmCustomOptions.solrdeployment.solr.options Custom Property

Parameters with Description

The following table describes the parameters for the *LdmCustomOptions.solrdeployment.solr.options* custom property that you can set in the Catalog Service:

Parameter	Description
MaxDirectMemorySize	Maximum total size of java.nio (New I/O package) direct buffer allocations.
UseLargePages	Optimizes the processor translation lookaside buffers.
-Dsolr.hdfs.blockcache.slab.count	Number of memory slabs for allocation.
-Dsolrconfig.updatehandler.autocommit.maxdocs*	Maximum number of Solr documents stored in memory before performing an auto commit.
*the default configuration *LdmCustomOptions.solrdeployment.solr.options=-Dsolrconfig.updatehandler.autocommit.maxdocs=100000* indicates that Enterprise Data Catalog performs an auto commit every time after storing 100000 Solr documents in memory.	

Solr Deployment Parameters Based on Data Size

The following table lists the Solr deployment parameters and their values based on the data size:

Data Size	Values
Low	LdmCustomOptions.solrdeployment.solr.options=-XX:MaxDirectMemorySize=3g -XX:+UseLargePages -Dsolr.hdfs.blockcache.slab.count=24 -Dsolrconfig.updatehandler.autocommit.maxdocs=100000
Medium	LdmCustomOptions.solrdeployment.solr.options=-XX:MaxDirectMemorySize=5g -XX:+UseLargePages -Dsolr.hdfs.blockcache.slab.count=38 -Dsolrconfig.updatehandler.autocommit.maxdocs=100000
High	LdmCustomOptions.solrdeployment.solr.options=-XX:MaxDirectMemorySize=10g -XX:+UseLargePages -Dsolr.hdfs.blockcache.slab.count=76 -Dsolrconfig.updatehandler.autocommit.maxdocs=100000

LdmCustomOptions.SolrNodeProperties

Parameters with Default Values

The following table lists the *LdmCustomOptions.SolrNodeProperties* parameters and their values based on the data size in the `custom.properties.<loadtype>` files:

Option	Low	Medium	High
yarn.component.instances	1	3	6
yarn.memory	2560	2560	2560
yarn.vcores	1	1	1
xmx_val	2048m	2048m	2048m
xms_val	512m	512m	512m
solr.replicationFactor	1	1	1
solr.numShards	1	3	6

Solr Node Properties Based on Data Size

The following table lists the parameters and their default values based on the data size for Solr node properties:

Data Size	Values
Low	LdmCustomOptions.SolrNodeProperties=yarn.component.instances=1,yarn.memory=2560,yarn.vcores=1,xmx_val=2048m,xms_val= 512m,solr.replicationFactor=1,solr.numShards=1
Medium	LdmCustomOptions.SolrNodeProperties=yarn.component.instances=3,yarn.memory=2560,yarn.vcores=1,xmx_val=2048m,xms_val= 512m,solr.replicationFactor=1,solr.numShards=3
High	LdmCustomOptions.SolrNodeProperties=yarn.component.instances=6,yarn.memory=2560,yarn.vcores=1,xmx_val=2048m,xms_val= 512m,solr.replicationFactor=1,solr.numShards=6

Approximate Size of Index Files

Low. 800 MB

Medium. 16 GB

High. 40 GB

Note:

- The configuration shown for the low data size caches the whole index.
- The configuration shown for the medium data size assumes that you are running HDFS and Apache Solr on a host with 32 GB of unused memory.

Scaling Up the Number of Search Users

The High load type can accommodate 100 concurrent search users with default *solr.numShards=6* parameter. You can increase the number of search users by increasing the number of Solr shards. To support an additional 25 search

users, you can increase the *solr.numShards* value for High load type by 1 which increases the number of cores usage by 8 with an additional memory usage of 5 GB. Re-index the catalog after you increase the *solr.numShards* value.

For example, to add an additional 50 search users to the existing 100 users, set the *solr.numShards* parameter to 8. When you increase the Solr shards value to 8, the CPU cores requirement increases by 16 cores in addition to the existing number of cores, and the memory requirement increases by 10 GB in addition to the existing memory usage. After you increase the Solr shards value, re-index the catalog.

Enterprise Data Catalog Agent

You can use Enterprise Data Catalog Agent to extract metadata from data sources that run on Microsoft Windows.

The following table lists the minimum system requirements to deploy Enterprise Data Catalog Agent:

Component	Minimum System Requirements
CPU Cores	8
Memory	16 GB

Log Files

To monitor the performance and to identify issues, you can view the log files. Log files are generated at every stage of metadata extraction and ingestion.

You can view the following log files for Enterprise Data Catalog:

Catalog Service Log File

Navigate to the `<$INFA_HOME>/logs/<node_name>/services/CatalogService/<Catalog_service_name>/LDM.log` directory to view the Catalog Service log file.

Scanner Log File

In Catalog Administrator, you can configure the scanner log level for a resource and MITI log level for data sources that support MITI. You can set the log level as NONE, FATAL, ERROR, WARNING, STATUS, INFO, or DEBUG.

To configure the log level, perform the following steps:

- To configure the scanner log level, navigate to the resource and set the following JVM option in the **Metadata Load Settings** > **Advanced Properties** section:
  `-Dscannerloglevel=<scanner_log_level>`

  For example, `-Dscannerloglevel=INFO`
- To configure the MITI log level, navigate to the resource and set the following JVM option in the **Metadata Load Settings** > **Advanced Properties** section:
  `-DagentLogLevel=<scanner_log_level>`

  For example, `-DagentLogLevel=INFO`.

Note: By default, the scanner log level is INFO and log level for data sources supported by MITI is DEBUG. It is recommended to reduce the log level when the log file size increases in GB during a single large resource run.

In Catalog Administrator, you can view the Log URL for resources in the **Monitoring** > **Tasks** section.

Use the Yarn Resource Manager URL in the cluster to open the Resource Manager to view the following messages and log file:

- To view the stdout messages, click the **scanner <application id> URL** > **appattempt_<application id>_000001 URL** > **container_<application id>_01_000002 URL** > **Logs URL**.
- To view the Scanner.log file, navigate to the **yarn logs** > **applicationId application_<Scanner application Id>** section.

Ingestion Log File

Use the Yarn Resource Manager URL to open the Resource Manager to view the following messages and log file:

- To view the stderr messages, click the **ingestion <application id> URL** > **Logs URL**.
- To view the Ingestion.log file, navigate to the **yarn logs** > **applicationId application_<Ingestion application Id>** section.

HBase Log File

Use the Yarn Resource Manager URL to open the Resource Manager to view the following log files:

- To view the log files, click **Hbase <application id> URL** > **appattempt _<application id>_000001 URL** > **container_<application id>_02_<000002> [multiple Hbase Region server container] URL** > **Logs URL**.
- To view the Hbase.log file, navigate to the **yarn logs** > **applicationId application_<Hbase application Id>** section.

Solr Log File

Use the Yarn Resource Manager URL to open the Resource Manager to view the following log files:

- To view multiple log files, click **solr <application id> URL** > **appattempt _<application id>_000001 URL** > **container_<application id>_01_<000002> [multiple solr container] URL** > **Logs URL**.
- To view the Solr.log file, navigate to the **yarn logs** > **applicationId application_< Solr application Id>** section.

Data Integration Service Log File

To view the log file, navigate to the `<$INFA_HOME>/logs/<node_name>/services/DataIntegrationService/disLogs/dis_<dis_ service_name>_<node_name>.log.0` directory.

Mapping Log File

To view the log file, navigate to the `<$INFA_HOME>/logs/<node_name>/services/DataIntegrationService/disLogs/profiling/<logs>` directory.

Performance Tuning Parameters for Profiling

Tuning profiling performance involves configuring the Data Integration Service parameters, Data Integration Service concurrency parameters, and profiling warehouse parameters.

Tuning Data Integration Service Parameters for Profiling

To optimize data profiling, you can configure Data Integration Service parameters, Data Integration Service profiling tuning parameters, and Data Integration Service grid configuration. You can also tune unique key inference performance parameters.

You need to configure the Temporary Directories and Maximum Execution Pool Size parameters for the Data Integration Service. You can configure parameters, such as Reserved Profile Threads that apply to the Profiling Service Module. Before you use the parameter recommendations, verify that you have identified a node or grid and the requirement is to configure the node or grid optimally to run profiles.

The following configuration parameters have an impact on different components of the profiling and discovery installation. You can increase or decrease the values for these parameters based on the performance requirements:

Parameter	Description
Maximum Execution Pool Size	The maximum number of requests that the Data Integration Service can run concurrently. Requests include data previews, mappings, and profiling jobs. This parameter has an impact on the Data Integration Service.
Maximum Profile Execution Pool Size	The total number threads to run profiles. This parameter has an impact on the Profiling Service Module.
Temporary Directories	Location of temporary directories for the Data Integration Service process on the node. This parameter has an impact on the Data Integration Service machine.

Note: For more information about Data Integration Service parameters, see the *Profiling and Discovery Sizing Guidelines* article.

Data Integration Service Profiling Tuning Parameters

The following table lists the values for the Data Integration Service profiling tuning parameters:

Tuning Parameter	Value for 100,000 or lesser Rows with Row Sampling Enabled	Value for 100,000 or more Rows with Row Sampling Disabled	Description
Maximum Concurrent Columns	100	5	Maximum number of columns merged in single mapping.
Maximum Column Heap Size	512	512	Maximum cache size allocated for a mapping.
AdvancedProfilingServiceOptions.ColumnsThreshold	100	Default	Maximum number of columns merged for a mapping. Applicable only for column profiling.
AdvancedProfilingServiceOptions.DomainMappingColumnThreshold	100	50	Maximum number of columns merged for a mapping. Applicable only for data domain discovery.
AdvancedProfilingServiceOptions.MaxDataDomains	150	Default	Maximum number of data domains combined for a data domain discovery mapping.
ExecutionContextOptions.Optimizer.MaxMappingPorts	-1	Default	Controls platform optimization during mapping execution.
ExecutionContextOptions.Optimizer.PushdownType	none	Default	Controls pushdown optimization during mapping execution.

Tuning Parameter	Value for 100,000 or lesser Rows with Row Sampling Enabled	Value for 100,000 or more Rows with Row Sampling Disabled	Description
ExecutionContextOptions.Optimizer.ExtractSourceConstraints	false	Default	Controls extracting constraint details for a relational source during mapping execution.
ExecutionContextOptions.JVMOption1	1024M	1024M	Controls mapping process heap size.
ExecutionOptions.MaxProcessLifeTime	1800000	Default	Controls mapping process life time.
ExecutionContextOptions.Optimizer.NativeExpressionTx	false	Default	Controls SISR optimization during mapping execution.

Data Integration Service Node Scaling for Profiling

The execution time and throughput for data discovery varies based on the following parameters. It also depends on the hardware configuration of the Data Integration Service node.

- Volume of data in the tables and files
- Number of rows
- Number of unique or duplicate data sets
- Number of columns
- Column length
- Sampling options
- Number of data domains

You can choose a configuration from the following options to scale the Data Integration Service node for profiling:

- **Option 1 — Vertical hardware scaling with a standalone node**
  - This option is recommended if you have a single profiling resource.
  - The throughput for a single profiling resource can be increased by vertically scaling the node hardware configuration.
  - To achieve vertical scaling, you must calculate the Data Integration Service concurrency parameters for profiling and increase the number of CPU cores and system memory on the node. For information on how to calculate the Data Integration Service concurrency parameters, refer to the "Data Integration Service Concurrency Parameters" on page 28.
- **Option 2 — Horizontal hardware scaling with multiple nodes**
  - This option is recommended if you have multiple profiling resources.
  - The throughput for multiple profiling resources can be increased by horizontally scaling the node hardware configuration. This configuration is used to increase the throughput across resources.
  - Create multiple nodes with dedicated profiling warehouse schema that can be selected at the resource level under the domain connection settings.

- **Option 3 — Grid configuration**
  - This option is used to increase the throughput for a single profiling resource by horizontal scaling.
  - Using this configuration, you can improve the data discovery throughput by up to 1.4x by increasing the total hardware capacity by 2x across multiple grid nodes.
  - The generation of mappings and compilation is performed on the master node. The execution takes place on both the master and the secondary nodes. Therefore, the hardware utilization on the master node is always higher than the secondary nodes.
  - The minimum hardware requirement recommended for a node used in a grid configuration is 32 CPU cores and 64 GB memory.
  - For information on how to configure multiple nodes in a grid configuration, refer to the "Data Integration Service Grid Configuration for Enterprise Data Catalog" on page 27 section.

Data Integration Service Grid Configuration for Enterprise Data Catalog

In Enterprise Data Catalog, to use the Data Integration Service on a grid, you need to configure a shared directory with read and write access on all the nodes in the grid.

To configure the *<INFA_HOME>/<profile_temp_folder>* shared directory on all the nodes, perform the following steps:

1. Log in to Informatica Administrator.
2. Navigate to **Manage** > **Services and Nodes** > **Domain Navigator**.
3. Select the grid in the **Domain Navigator**.
4. In the **Properties** > **Custom Properties** section, click the **Edit** icon.
5. In the **Edit Custom Properties** dialog box, click **New**.
6. In the **New Custom Property** dialog box, enter the following details:
   a. In the **Name** field, enter **AdvancedProfilingServiceOptions.CustomFlag8**.
   b. In the **Value** field, enter the temporary folder. For example, `/data2/profile_temp`.
7. Recycle the Data Integration Service.

In Enterprise Data Catalog, you can use the shared directory to perform the following tasks:

- To stage XML or JSON schemas on a Data Integration Service node
- To stage unstructured files and results on a Data Integration Service node
- To stage similarity results on a Data Integration Service node

Tuning Performance for Unique Key Inference Process

The unique key inference throughput and execution time are exponential and they depend on the following factors:

- Schema size
- Data set volume
- Profiling sample size
- Profiling concurrency

To tune unique key inference performance, configure the **ExecutionContextOptions.JVMoption1 = -Xmx4096M** JVM option in the **Informatica Administrator** > **Manage** > **Services and Nodes** > **Data Integration Service** > **Custom Properties** section.

Data Integration Service Concurrency Parameters

To increase the profiling performance throughput, you can tune the following Data Integration Service concurrency parameters:

- Maximum On-Demand Execution Pool Size
- Maximum Profile Execution Pool Size
- Maximum Concurrent Profile Jobs
- Maximum Concurrent Profile Threads
- AdvancedProfilingServiceOptions.FileConnectionLimit. This parameter applies to SAP, XML, and JSON sources.
- AdvancedProfilingServiceOptions.HiveConnectionLimit. This parameter applies to Hive sources.
- Maximum # of Connections for source connection

You can configure the concurrency parameters in one of the following ways:

- In the Administrator tool, navigate to the Data Integration Service properties to configure the concurrency parameters.
- In the command prompt, navigate to the `<INFA_HOME>/isp/bin/plugins/ps/concurrencyParameters/` location.
  The *ConcurrencyParameters.properties* file in this location contains the default values of the Data Integration Service concurrency parameters for each load type. Run the *setConcurrencyParameters* script with the load type option to update the parameter values simultaneously for Data Integration Service.

Calculate Concurrency Parameter Values

You can calculate the value for concurrency parameters based on the number of cores. It is recommended that you multiply the number of cores with 0.6 and round off the value to the nearest whole value. For example, if the number of cores configured for the Data Integration Service is 16, then the concurrency parameter value is calculated as follows: 16*0.6=9.6. The concurrency parameter value is rounded off to the nearest whole value, namely, 10.

The following table describes how to calculate the Data Integration Service concurrency parameters for profiling based on the number of cores:

Concurrency Parameter	Value*	Description
Maximum On-Demand Execution Pool Size	Number of cores x 0.6	Maximum parallel mappings run by the Data Integration Service that includes all types of mappings.
Maximum Profile Execution Pool Size	Number of cores x 0.6	Maximum parallel profiling mappings run by the Data Integration Service.
Maximum Concurrent Profile Jobs	Number of cores x 0.6	Maximum parallel tables or files processed.
Maximum Concurrent Profile Threads	Number of cores x 0.6	Maximum parallel mappings run by the Data Integration Service for File sources.
AdvancedProfilingServiceOptions.FileConnectionLimit (For CSV, XML, and JSON metadata sources)	Number of cores x 0.6	Maximum parallel profiling mappings run by the Data Integration Service for File sources.

Concurrency Parameter	Value*	Description
AdvancedProfilingServiceOptions.HiveConnectionLimit (For Hive metadata sources)	Number of cores x 0.6	Maximum parallel profiling mappings run by the Data Integration Service for Hive sources.
Maximum # of Connections	Number of cores x 0.6	Total number of database connections for each resource.
* Indicates the value of the parameter if the number of rows is less than or greater than 100, 000 rows with sampling of rows enabled or disabled.		

setConcurrencyParameters Script

You can run the script to update all the parameter values in the *ConcurrencyParameters.properties* file simultaneously. After you run the script, the script stops the Data Integration Service, updates the parameter values in the Data Integration Service, and then enables the Data Integration Service.

The *Maximum # Connection Pool Size* parameter is configured at the data source level. Therefore, you need to update this parameter in the Administrator tool for relational sources.

The setConcurrencyParameters script uses the following syntax:

```
setConcurrencyParameters.sh
[small/medium/large]
[DomainName]
[DISName]
[Username]
[Password]
```

The following table describes the setConcurrencyParameters script arguments:

Argument	Description
small/medium/large	Required. Enter small, medium, or large.
DomainName	Required. Enter the domain name.
DISName	Required. Enter the Data Integration Service name.
Username	Required. Enter the user name that the Data Integration Service uses to access the Model Repository Service.
Password	Required. Enter the password for the Model repository user.

Usage

```
./setConcurrencyParameters.sh [small/medium/large] [DomainName] [DISName] [Username]
[Password]
```

Example

```
./setConcurrencyParameters.sh small domain DIS Administrator Administrator
```

Default Parameter Values for Small, Medium, and Large Options in the Script

Parameter	Small	Medium	Large
Data Integration Service server hardware	CPU cores is16. Memory is 32 GB.	CPU cores is 32. Memory is 64 GB.	CPU cores is 64. Memory is 128 GB.
Maximum On-Demand Execution Pool Size	10	20	40
Maximum Profile Execution Pool Size	10	20	40
Maximum Concurrent Profile Jobs	10	20	40
Maximum Concurrent Profile Threads	10	20	40
AdvancedProfilingServiceOptions.FileConnectionLimit (For SAP, XML, JSON sources)	10	20	40
AdvancedProfilingServiceOptions.HiveConnectionLimit (For Hive sources)	10	20	40
Maximum # of Connections for source connection	10	20	40

Tuning Profiling Warehouse Parameters

The profiling warehouse stores profiling results. More than one Profiling Service Module may point to the same profiling warehouse. The main resource for the profiling warehouse is disk space.

The following table lists the parameters that you need to tune to improve the performance of the profiling warehouse based on the size of data:

Parameter	Low	Medium	High	Description
Tablespace	50 GB	200 GB	500 GB	The maximum amount of disk space required to temporarily store the profiling results before these results get stored in the Enterprise Data Catalog.
CPU	4	8	16	The number of cores required by the profiling warehouse

Note: For more information about the profiling warehouse guidelines, see the Profiling and Discovery Sizing Guidelines article.

Tuning Performance for Profiling on Spark Engine

To increase the profiling throughput for high volume sources, run the profiles on the Spark engine in the Hadoop environment.

To optimize profiling performance, run the following resources on the Spark engine:

- HDFS sources
- Hive sources
- Partitioned parquet files on Amazon S3 and Microsoft Azure Data Lake Store Gen2.

To tune the profiling throughput, configure the following parameters:

- Increase the value of **spark.yarn.executor.memoryOverhead** from 384 to 2048.
- Configure the **spark.dynamicAllocation.maxExecutors** parameter to 5 for every 3 nodes. For example, you can configure **spark.dynamicAllocation.maxExecutors** to 10 for 6 nodes.

To configure the parameters, navigate to the **Informatica Administrator** > **Manage** > **Connections** > **Hadoop CCO connection** > **Spark Configuration** > **Advanced Properties** section.

For information about best practices and to tune the Spark engine, see the Performance Tuning and Sizing Guidelines for Informatica® Data Engineering Integration 10.4.0 and Performance Tuning for the Spark Engine articles.

Tuning Performance for Profiling on Blaze Engine

To increase the profiling throughput for high volume sources, you can run the profiles on the Blaze engine in the Hadoop environment for relational sources that exceed 100 million rows.

The following table lists the custom properties for the Data Integration Service and the values you need to configure to run the profiles on the Blaze engine:

Parameter	Value
ExecutionContextOptions.Blaze.AllowSortedPartitionMergeTx	false
ExecutionContextOptions.GridExecutor.EnableMapSideAgg	true
ExecutionContextOptions.RAPartitioning.AutoPartitionResult	true
AdvancedProfilingServiceOptions.BlazeBatchSize	1
AdvancedProfilingServiceOptions.DomainMappingColumnThreshold	1
AdvancedProfilingServiceOptions.ColumnsThreshold	1

Note: When you run profiles on the Blaze engine, the profiling service stages the source data in the cluster nodes which uses higher disk space than the source size.

For information about best practices and to tune the Blaze engine, see the *Performance Tuning and Sizing Guidelines for Informatica Big Data Management* and *Best Practices for Big Data Management Blaze Engine* articles.

Memory Sizing to Run Profiles on Large Unstructured Files

You can run a profile on unstructured files, such as PDF, Microsoft Excel, and Microsoft Word files.

Memory Requirements to Run Profiles on Large PDF Data Sources

Large unstructured files in PDF format require more memory during profiling.

The following table lists the memory that you can configure for the *ExecutionContextOptions.JVMOption1* parameter based on the PDF file size:

PDF File Size	Memory (in MB)
Less than or equal to 25 MB	1024M
Less than or equal to 50 MB	2048M
Less than or equal to 100 MB	5120M

Memory Requirements to Run Profiles on Large Microsoft Excel or Microsoft Word Files

The following table lists the memory that you can configure for the *ExecutionContextOptions.JVMOption1* parameter based on the Microsoft Excel or Microsoft Word file size:

File Size	Memory (in MB)
Less than or equal to 5 MB	1024M
Less than or equal to 25 MB	2048M
Less than or equal to 100 MB	7168M

To tune the *ExecutionContextOptions.JVMOption1* parameter, navigate to the **Manage** > **Services and Nodes** > **Data Integration Service** > **Custom Properties** section in Informatica Administrator.

Tuning Resource Configuration Parameters for Profiling

Snowflake Resource

To optimize the profiling performance for Snowflake high volume resources, configure the following custom properties in Informatica Administrator:

- **AdvancedProfilingServiceOptions.ColumnsThreshold**. Set the value to **10**.
- **ExecutionContextOptions.JVMoption1**. Set the value to **-Xmx2048M**.

To configure these parameters, navigate to the **Services and Nodes** > **Data Integration Service** > **Custom Properties** section.

Microsoft Azure Data Lake Store Gen2 Resource

You can configure the following parameters to optimize the profiling performance for Microsoft Azure Data Lake Store Gen2 resources:

Heap Size for pmdtmsvc2 Process

To set the maximum heap size for pmdtmsvc2 process used for profiling Microsoft Azure Data Lake Store Gen2 high volume resources in the native runtime environment, set the following custom property :

`ExecutionContextOptions.JVMoption1` custom property to -Xmx4096M value.

The pmdtmsvc2 process memory requirement might increase for a large file. When the configured heap size does not meet the file processing memory requirement, a heap size error appears and the mapping is marked as failed.

To configure the custom property, navigate to the **Informatica Administrator** > **Services and Nodes** > **Data Integration Service** > **Custom Properties** section.

Concurrent Threads

You can set concurrent threads to download the files from Microsoft Azure Data Lake Store Gen2 source. By default, this option is set to 10. It is recommended to configure this value in the range of current number of cores to twice the number of CPU cores. For example, on a 16 CPU cores server, you can configure this parameter from 16 through 32.

You can configure the following option at resource level to override the default value:

```
-Dconnector.access.concurrent.thread=<value>
```

To configure this JVM option at resource level in Catalog Administrator, open the resource and set the JVM option on the **Metadata Load Settings** tab.

Block Size

You can set the block size to download the files in chunks using multiple threads. By default, the option is set to 8MB.

You can configure this option at resource level to override the default value:

```
-Dconnector.access.block.sizeInMB=<value>
```

It is recommended that you consider the pmdtmsvc2 process heap size and concurrent threads to configure this option to an optimum value to avoid redundant network calls.

To configure this JVM option at resource level in Catalog Administrator, open the resource and set the JVM option on the **Metadata Load Settings** tab.

Heap Size

Configure the Data Integration Service heap size to 10GB in Informatica Administrator before you run the data domain discovery profiles in Catalog Administrator. You can increase this value if the data domains exceed 100.

Cassandra Resource

You can configure the following parameters to optimize the profiling performance for Cassandra resources:

Heap Size for pmdtmsvc2 Process

Configure the **ExecutionContextOptions.JVMoption1** custom property to -Xmx2048M to optimize the profiling performance for Cassandra high volume resources. This parameter sets the maximum heap size for pmdtmsvc2 process used for profiling.

To configure these parameters, navigate to **Informatica Administrator** > **Services and Nodes** > **Data Integration Service** > **Properties** > **Custom Properties** section.

Heap Size

Configure the Data Integration Service heap size to 10 GB when you run data domain discovery on Cassandra resources. Increase the heap size if the number of data domains exceed 100.

Performance Tuning Parameters Definitions

To optimize the performance of Enterprise Data Catalog, you can configure the metadata ingestion parameters, Apache HBase database tuning parameters, and Apache Solr parameters.

Parameters for Tuning Metadata Ingestion

The following table lists the metadata ingestion parameters that you can configure in Enterprise Data Catalog to tune ingestion performance:

Parameter	Description
driver-memory	Memory required by the Spark driver.
executor-memory	Memory required by Spark executors.
num-executors	Number of executors spawned by Spark to process the metadata.
executor-core	Number of CPU cores allocated for each executor.
driver-java-options=-XX:MaxPermSize	Perm gen size for the Spark driver.
conf Spark.executor.extraJavaOptions=-XX:PermSize	Perm gen size for the Spark executor.
conf Spark.storage.memoryFraction	Fraction of java heap space used for Spark-specific operations such as aggregation.
ingest.batch.delay.ms	Represents clock skew across nodes of the cluster. Default is two minutes and the value must be set higher than the skew.
ingest.max.ingest.xdocs.int	This parameter is deprecated. Maximum number of documents to be processed in one batch.
ingest.max.ingest.facts.int	Maximum amount of facts about an object that can be processed in single batch of ingestion.
ingest.batch.time.ms	All the documents remaining from the previous batch processed and the present batch. These documents get processed with the next batch. This value is restricted to the batch size specified earlier.
ingest.partitions.int	Number of partitions to be used for ingestion.
ingest.propagation.partitions.int	Number of partitions to be used for propagation.
ingest.sleep.time.ms	Interval between different ingestions.
ingest.xdoc.memory.buffer.bytes	Total amount of memory that can be used by XDOCS. **Note:** If you increase the value of this parameter, make sure that you increase the value for the metadata extraction scanner memory parameter. Failing to increase the scanner memory parameter might result in out of memory errors.
titan.ids.num-partitions	Used by Titan to generate random partitions of the ID space and helps avoiding region-server hotspot. This value must be equal to the number of region servers.
titan.cluster.max-partitions	Determines the maximum number of virtual partitions that Titan creates. This value must be provided in multiples of two.
titan.storage.hbase.region-count	Number of HBase regions to be used for the TITAN table.

Parameter	Description
HbaseTableSplits	Controls the number of pre-splits in the Hbase table.
ingest.phoenix.query.timeoutMs	Amount of time within which the HBase operation must complete before failing.

Parameters for Tuning Apache HBase

The Apache HBase tuning parameters include parameters for the Apache HBase site configuration properties, the HBase region server properties, the HBase master properties, and the HBase Slider Master App properties.

The following tables list the parameters for tuning different properties of Apache HBase:

HBase-site Configuration Properties

The following table lists the HBase-site configuration properties:

Parameter	Description
hbase_master_heapsize	Heap size for HBase master.
hbase_regionserver_heapsize	Heap space for region server.
hbase.master.handler.count	Total number of RPC instances on HBase master to serve client requests.
hbase.regionserver.handler.count	Handler count for the region server.
hbase.hstore.blockingStoreFiles	Number of store files used by HBase before the flush is blocked.
hbase.hregion.majorcompaction	Time between major compactions of all the store files in a region. Setting this parameter to 0 disables time-based compaction.

HBase Region Server Properties

The following table lists the HBase region server properties:

Parameter	Description
yarn.component.instances	Number of instances of each component for slider. This parameter specifies the number of region servers that are run.
yarn.memory	Amount of memory allocated for the container hosting the region server.
yarn.vcores	Number of cores allocated for the region server.

HBase Master Properties

The following table lists the HBase Master properties:

Parameter	Description
yarn.component.instances	Number of instances of each component for slider. This parameter specifies the number of master servers that are run.
yarn.memory	Amount of memory allocated for the container hosting the master server.
yarn.vcores	Number of cores allocated for the master server.

HBase Slider Application Master Properties

The following table lists the HBase Slider Application Master properties:

Parameter	Description
jvm.heapsize	Amount of memory allocated for the container hosting the master server.
yarn.component.instances	Number of instances of each component for slider. This parameter specifies the number of master servers that are run.
yarn.memory	Number of cores allocated for the master server.
yarn.vcores	Memory used by the slider app master.

Parameters for Tuning Apache Solr

These parameters include the Apache Solr Slider app master properties and the Enterprise Data Catalog custom options for the Apache Solr node. The following tables list the Apache Solr parameters that you can use to tune the performance of search in Enterprise Data Catalog:

Apache Solr Slider App Master Properties

Parameter	Description
jvm.heapsize	Memory used by the slider app master.
yarn.component.instances	Number of instances of each component for slider. This parameter specifies the number of master servers that are run.
yarn.memory	Amount of memory allocated for the container hosting the master server.
yarn.vcores	Number of cores allocated for the master server.

Enterprise Data Catalog Custom Options Apache Solr Node Properties

Parameter	Description
xmx_val*	Solr maximum heap
xms_val	Solr minimum heap
yarn.component.instances	Number of instances of each component for slider. This parameter specifies the number of master servers that are run.
yarn.memory	Amount of memory allocated for the container hosting the master server.
yarn.vcores	Number of cores allocated for the master server.
solr.replicationFactor	Number of Solr index replications.
solr.numShards	Number of Solr shards.

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